SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

16th Floor, BoC Merchant Tower, No. 28, St. Michael's Road, Colombo 03

PARTICIPANT INFORMATION SHEET

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery (CareSense)

Principal Investigator: Prof. Samantha Thelijjagoda, Senior Professor, Department of Computer Systems Engineering, Faculty of Computing, SLIIT

Researchers: Dayana Priyadharshani Kumar (IT22178640), Harishalinee Elangovan (IT22057488), Vidursha Prabagaran (IT22294098), Kaushalya Nagenthraraja (IT22289384) — Faculty of Computing, SLIIT

1. Invitation

You are being invited to take part in a research project being carried out as part of the BSc Special (Hons) Research Project module at the Faculty of Computing, SLIIT. Before you decide whether to participate, it is important that you understand why the research is being done and what it will involve. Please take time to read the following information carefully.

2. What is the purpose of this project?

This project is developing an AI-based system, called CareSense, to help caregivers monitor and support the people they care for. One part of this system automatically detects fall risk from ordinary webcam video. We are asking for your help to test this system with a short video recording of a simulated fall or movement scenario, so we can check that it works correctly in a realistic home setting.

3. Why have I been invited?

You have been personally invited by the researcher (a family member/close contact) to help test the system, since this stage of testing does not use the general public and instead relies on a small number of known, consenting volunteers.

4. What will I be asked to do?

- You will be asked to perform a short, simple movement or simulated fall scenario (e.g., sitting down, standing up, or lying down safely) in front of a webcam in a home setting.
- The recording will last only a few minutes.
- No physical risk activity (e.g., an actual uncontrolled fall) will be requested — all scenarios are simulated safely and slowly.
- The recording will be used only to test whether the system correctly detects the simulated scenario — it will not be used to train the underlying AI model.

5. Do I have to take part?

No. Participation is entirely voluntary. You are free to withdraw at any time, without giving a reason, and without any consequence. If you withdraw, any recording of you will be deleted.

6. What will happen to the information I provide?

- The video recording will be stored securely on a password-protected device accessible only to the research team.

- Your recording will be used only for internal testing of the fall-detection system, not published or shared publicly in identifiable form.
- Any results or footage used in reports/presentations will be described only in general terms (e.g., “a test recording of a simulated fall scenario”) without your name or other identifying information, unless you separately agree otherwise.
- The recording will be permanently deleted from all devices after the project is completed and graded.

7. Are there any risks or benefits to taking part?

There are no significant physical or psychological risks, since all scenarios are simulated safely under your control. There is no direct personal benefit to you, but your participation will help the research team validate a system intended to improve caregiver support and safety monitoring.

8. Who has reviewed this project?

This project has been reviewed as part of the SLIIT Faculty of Computing ethics review process.

9. Contact for further information

If you have any questions about this project, please contact any of the research team members:

Dayana Priyadharshani Kumar — it22178640@my.sliit.lk | 0764043970

Harishalinee Elangovan — it22057488@my.sliit.lk | 0775073188

Vidursha Prabagaran — it22294098@my.sliit.lk | 0779890702

Kaushalya Nagenthraraja — it22289384@my.sliit.lk | 0764232082

Supervisor: Prof. Samantha Thelijjagoda — samantha.t@sliit.lk | 077-440-2550

Thank you for considering taking part in this research.